Lab 6 · Case Study: A Cluster on the Surgical Ward

Day 2 · 13:15–15:00 · 105 min · Groups of 4 Objective: LO3–LO5 — Combine genomic and epidemiological evidence to decide whether an outbreak is happening, and communicate the decision

> ⚠️ **About this case:** the genomes are **real**: MRSA isolates from a published hospital outbreak investigation. The hospitals, wards, patients and dates are **fictional**, written so the case feels like our setting. The real story is revealed at the end.

The situation

District Hospital A (fictional) · Monday, 16 March 2026

The infection prevention and control (IPC) nurse writes to you:

> "Since early February we've had **more MRSA wound infections than usual on Surgical Ward B**. The surgeons say it's bad luck. Theatre staff say it's the ward. The ward says it's the theatre. The hospital director asks whether we should **close Ward B to admissions**. Closing it would cancel about 40 operations a week. The lab stored all MRSA isolates since September, and we got genomes back from our sequencing partner last week. **Can you tell us what's going on by Wednesday's IPC meeting?**"

Your evidence

1. **Microreact project** from Lab 5 (tree + timeline + metadata), or the `koser_tree.nwk` and `lab6_metadata.csv` files
2. **SNP distance matrix**: `koser_snp_matrix.tsv` (in the shared Galaxy history). Open it in Galaxy (👁) or download it and open it in Excel.
3. **Pathogenwatch results** from Lab 5 (ST and AMR per isolate)
4. **Metadata table** (`lab6_metadata.csv`): patient code, ward, procedure, theatre, surgery date, sample date

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Group roles (rotate if time allows)

Role	Job
Genomics lead	Reads the tree and SNP matrix
Epidemiologist	Builds the timeline: who was where, when
Clinician / IPC	Thinks through transmission routes and actions
Communicator	Writes and delivers the 5-minute briefing

Part A — Describe the cases (15 min)

Using the metadata, build an **epidemic curve**: sketch it on paper, with weeks along the x-axis and the number of cases on the y-axis.

Week starting	Ward B cases	Other ward/hospital cases

✍️ Questions:

1. How many MRSA isolates in total? How many from Ward B since 1 February?
2. Do **all** Ward B cases look the same from the **lab** report (species + cefoxitin)? Does that mean they are one outbreak?

Part B — What does the genome say? (25 min)

B1. Sequence types

From Pathogenwatch, list each isolate's ST. Group the isolates by ST.

B2. SNP distances

Open the SNP matrix. Each cell is the number of single-letter differences between two isolates.

- Highlight all pairs with **≤ 30 SNPs** (our working threshold for possible recent *S. aureus* transmission; published thresholds vary, see L4)
- Which isolates form a **cluster** (all closely related to each other)?

Cluster	Isolates (patient codes)	SNP range within cluster	Closest outside isolate (SNPs)

B3. Tree

In Microreact, colour the tree by **ward**, then by **theatre**. Does the cluster from B2 appear as a tight group on short branches?

Part C — Put genomics and epidemiology together (20 min)

For **each** Ward B case since February, decide:

Patient	In genomic cluster?	Epi link (ward/theatre/ time overlap)?	Conclusion: **linked / not linked / uncertain**

✍️ Discussion questions:

1. Is there an outbreak? How confident are you?
2. Is there a Ward B patient who is **not** part of the cluster even though they were on the ward at the same time? What does that tell you?
3. Is there a cluster patient who was **not** on Ward B? What do they have in common with the others? → **What is the likely source or route?**
4. Do any isolates from **other hospitals** belong to the cluster?
5. What **other information** would you want? (e.g. staff screening, environmental swabs, theatre logs)

Part D — Prepare the briefing (25 min)

Prepare a **5-minute briefing** for the IPC committee (director, chief surgeon, nursing lead, lab head). They are **not** genomics experts.

Use this template (one flip-chart page or 3 slides):

> **1. Bottom line (1 sentence):** "We found / did not find evidence of an outbreak of _____ linked to _____." **2. What we did:** "We sequenced __ MRSA isolates from __ to __ and compared them genetically." **3. What we found:** one visual (screenshot of the Microreact tree + timeline) and 2–3 bullet points **4. What this means:** which cases are linked, and the likely route **5. Recommendations:** 3 concrete actions, with who does them and by when **6. What we don't know yet:** limitations and next steps

Rules: no jargon without explanation ("SNP" → "genetic differences"), and no more than **one** figure.

Part E — Briefings (15 min, plenary)

Each group gets **5 minutes** to present to the "IPC committee" (the instructor + one participant group acting as committee members). The committee must ask at least one question:

- "*Should we close Ward B?*"
- "*Is it the surgeons' fault?*"
- "*How sure are you?*"
- "*What will this cost?*"

The reveal (instructor, 5 min)

The genomes come from **Köser et al., *N Engl J Med* 2012**: an investigation of a suspected MRSA outbreak in a **special care baby unit** (neonatal unit) in Cambridge, UK. Rapid whole-genome sequencing distinguished the outbreak isolates from unrelated MRSA that looked identical on routine lab tests, and showed transmission that conventional methods had not resolved. A follow-up study (Harris et al., *Lancet Infect Dis* 2013) used genomics to trace transmission into the community and identify a **staff carrier**, which ended the outbreak.

Take-home messages:

1. **Same species + same antibiogram ≠ same outbreak.** Genomics separates linked from unlinked cases.
2. **Genomics finds hidden links** (shared theatre, staff, equipment) that ward-based thinking misses.
3. **Genomics can prevent unnecessary actions** (closing a ward) as well as trigger necessary ones.
4. Genomic results are only useful **with good metadata** and **clear communication**.

Genomic data: ENA project PRJEB2912 (Köser et al. 2012). Scenario metadata are fictional and created for teaching.